

Sailesh Dwivedy

Hilliard, OH | +1(650) 476-5385 | sdwivedy30@gmail.com | [LinkedIn](#) | [GitHub](#)
(Open to Relocate)

SUMMARY

AI/ML Engineer with 5+ years of experience building and deploying production ML, GenAI, and multi-agent systems across insurance, FinTech, healthcare, and research. Expertise in LangChain, LangGraph, LLM evaluation and observability, MLOps, and end-to-end ML pipelines, with a track record of delivering measurable automation, reliability, and business impact for Fortune 500 organizations.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, PyTorch, Scikit-learn, Pandas, NumPy
- **AI Agent & RAG:** LangChain, LangGraph, Deep Agents, Harness Engineering, Observability, Evaluation, Vector Stores
- **NLP & GenAI:** Hugging Face, Transformers, Fine-tuning, PEFT (LoRA/QLoRA), RLHF
- **Machine Learning & Deep Learning:** Supervised, Classification, Regression, Ensemble Models, Neural Networks
- **MLOps & Deployment:** Docker, FastAPI, Azure, GitHub, MLflow, Weights & Biases
- **Data Analysis & Visualization:** Exploratory Data Analysis (EDA), Tableau, Matplotlib

PUBLICATIONS, CERTIFICATIONS, AWARDS

- Logit-based losses limiting effectiveness of feature knowledge distillation, **NeurIPS Workshop 2025** [[Link](#)]
- AI Agents Fundamentals, Hugging Face, [[Certificate](#)]
- Generative AI with Large Language Models, Deep Learning.AI, and Amazon Web Services, [[Certificate](#)]
- Awarded as Best Performer of the team, *Associated with Higradius*

PROFESSIONAL EXPERIENCE

CENTRAL MUTUAL INSURANCE

ML Engineer

Ohio, USA

Mar 2026 - Present

Developing production multi-agent AI systems for claims leakage and summarization, transforming live claim, policy, and document data into source-grounded findings and liability reports with factual verification and human-in-the-loop review.

Claims Leakage Agent:

- **Architected and implemented a multi-agent AI system for Claims Leakage Auditing** using LangChain Deep Agents on LangGraph, reducing manual review time by **90%**, coordinating **8 concurrent specialist agents**.
- **Improved finding reliability and reduced hallucination** through **domain-aware actor-critic, source attribution, and human-in-the-loop verification**, validating outputs across **100 ground-truth claims**.
- Integrated **18 live claim and policy Model Context Protocol (MCP) tools** with **caching, retries, telemetry**.
- **Engineered an agent harness** with **Azure Foundry** models, **retries, timeouts, Pydantic-validated** structured outputs, **code-owned** logic, and **partial-result recovery** to keep long-running agent workflows resilient.
- **Engineered context management** for 200+ page policies using **tool-result offloading, Coverage Scout actor-critic preclassification, and shared-prefix prompt caching** to reduce context overhead, latency.
- **Built an evaluation and observability framework** using **OpenTelemetry, Azure Application Insights, and Foundry** for LLM-as-judge evaluation and trace-level monitoring of agent/tool latency, failure, token, and cost.
- **Containerized and deployed the Streamlit app** to on-prem **Kubernetes** and **Azure DevOps CI/CD**.

Claims Summarizer Agent:

- **Shipped a multi-agent Claims Summarizer architecture** using **Deep Agents** and **Azure OpenAI LLMs** to generate source-grounded, multi-section liability reports, **cutting manual report generation time by 83%**.
- Built **attachment retrieval workflows** with **human-in-the-loop selection**, filtering relevant medical records, demand letters, and other claim documents at request time to **optimize token usage**.
- Delivered a **Streamlit app** with **concurrent section-level actor-critic workflows**, cutting **runtime by 25%** while preserving factual consistency and enabling **feedback-driven revisions**.
- Developed an **agentic RAG chat bot** using **Azure AI Search** and **persistent chat memory**, supporting draft edits and **source-cited** answers across **5,000+ claim-related queries monthly**.

NEUROSPRING

AI/ML Engineer Intern

Remote (Napa, CA)

Jun 2025 - Mar 2026

Delivered ML and GenAI solutions in the healthcare domain, focusing on speech analysis and clinical text generation. Built end-to-end AI pipelines, integrating interpretability and retrieval systems to assist medical professionals.

- Achieved **80% F1** in slurred speech detection by building **ML pipelines** and **85% F1** using **deep learning** with **PyTorch**, tracked with **MLflow**, deployed via **FastAPI/Docker** on **GCP**.
- Integrated **explainability frameworks** through **SHAP** to provide transparent, interpretable AI outputs for clinical decision support, facilitating **stakeholder trust** and **regulatory compliance**.
- **Fine-tuned BioMistral-7B** using **LoRA** and **4-bit quantization**, in PyTorch/Hugging Face, enabling paramedics to generate clinical notes, tracked **50+ runs** with **Weights & Biases**.
- Developed a **Retrieval-Augmented Generation** pipeline to help paramedics access clinical guidelines.

IMAGE AND VIDEO COMPUTING LAB, UNIVERSITY OF COLORADO, BOULDER

Colorado, US

Volunteer Graduate AI/ML Researcher

May 2024 - Dec 2024

Improved CNN and ViT performance in image classification benchmarks by developing novel teacher-student optimization methods for knowledge distillation, resulting in faster convergence and higher accuracy.

- Co-authored **NeurIPS Workshop 2025** paper **Logit-Based Losses Limit the Effectiveness of Feature Knowledge Distillation**, improving CNN and ViT accuracy by **15%** with a novel logit-free KD framework.
- Achieved **80% faster convergence** on CIFAR/Tiny ImageNet by designing a novel **Knowledge Quality (KQ)** metric to automatically select optimal teacher layers, implemented in PyTorch.

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Odisha, India

Volunteer NLP Research Associate

Feb 2023 - Aug 2023

Conducted applied research on transformer-based document parsing to improve the accuracy of information extraction from business documents. The work demonstrated measurable accuracy gains over existing baselines.

- Evaluated **LayoutLMv3**, a **Transformer-based** document parser, against LayoutLMv2 and OCR-based baselines on document datasets, showing **5 to 10%** accuracy improvements in field extraction tasks.
- Tracked and compared **50+ experiments** with **MLflow**, monitoring loss convergence, and parameter tuning.

HIGHRADIUS TECHNOLOGIES PRIVATE LIMITED (FinTech)

Odisha, India

Associate Software Engineer - II (Data Science)

Jul 2019 - Feb 2023

Client: Red Bull, Danone, Hershey's, Staples, Pavion

Led a 10-member team delivering end-to-end production AI/ML solutions on AWS for 15+ Fortune 500 clients, improving revenue recovery and efficiency for fraud detection, payment prediction, cash forecasting.

- **Increased dollar recovery** by **12%** and **reduced manual research effort** by **25%**, building **classification models** (LightGBM, Random Forest) with **85% recall** for deduction fraud detection and **SHAP** interpretability.
- **Automated** low-dollar deduction **write-offs** and **prioritized research worklists** through AI action suggestions, accelerating dispute resolution and improving operational efficiency..
- **Increased on-time collections** by **17%** by building an **AI prioritization system** using payment history features to generate **personalized worklists**, enabling targeted follow-ups with high-risk customers.
- **Improved collector efficiency** by **20%** by predicting invoice **payment dates** and deriving a prioritization system with delay buckets using **regression models** (XGBoost, Random Forest) validated with **RMSE**.
- **Reduced model implementation** and **deployment timelines** by **40%** by automating preprocessing, feature engineering, training, and evaluation pipelines in **Amazon SageMaker**, with **MLflow** tracking.
- Collaborated with **client and finance teams** to drive adoption, and support decisions through **Tableau** insights.

PROJECTS

Logit-based losses limiting effectiveness of feature knowledge distillation, NeurIPS Workshop 2025 [\[Link\]](#)

Tech Stack: Python, PyTorch, CNNs, Vision Transformers (ViT), Knowledge Distillation, CIFAR, Tiny ImageNet

- Developed a **logit-free knowledge distillation framework** improving image-classification accuracy by **15%**.
- Designed a **Knowledge Quality (KQ) metric**, automating teacher-layer selection, for **80% faster convergence**.

ImageSense AI: Multimodal Image Retrieval System [\[Video Demo\]](#) | [Source Code](#)

Tech Stack: Python, CLIP, FAISS, LLaMA 3.2, Transformers,

- Developed **multimodal text-to-image retrieval pipeline** using CLIP-based image embeddings..
- Integrated LLaMA 3.2 via Ollama for semantic grounding and query expansion, improving search relevance.

EDUCATION

UNIVERSITY OF COLORADO BOULDER

Colorado, US

Master's in Computer Science (GPA: 3.90/4)

Aug 2023 – May 2025

KIIT UNIVERSITY

Odisha, India

Bachelor's in Computer Science

May 2016 – May 2020